# Annual Drinking Water Quality Report
## Vineyard City
2024

We are pleased to present to you, our customer, with this year's Drinking Water Quality report. This report is designed to inform you about the quality of the water and services we deliver to you every day. Our constant goal is to provide you with a safe and dependable supply of drinking water.

**We are pleased to report that our drinking water meets federal and state requirements.**
If you have any questions about this report or concerning your water utility, please contact Devan Peterson at 385-378-5197. We want our valued customers to be informed about their water utility. If you want to learn more, please attend any of our regularly scheduled meetings. They are held on the second and fourth Wednesday each month at 6 pm located in the city hall building.

The Vineyard City Water System provides its consumers with surface water. Our water sources are Orem City and Central Utah Water.

Corrosion of pipes, plumbing fittings and fixtures may cause metals, including lead and copper, to enter drinking water. To assess corrosion of lead and copper, Vineyard City conducts tap sampling for lead and copper at selected sites every three (3) years.

Vineyard City Water System has received a waiver from the Utah Division of Drinking in regard to the initial lead service line inventory. This wavier was granted because the water systems distribution lines are newer and were installed on or after 2016. If you have any question in regard to Vineyard City's distribution line inventory, Please contact Devan Peterson at 385-378-5197.

**TOTAL COLIFORM POSITIVE RESULTS:**
Water samples taken in October and November 2024 confirmed the presence of total coliform bacteria. Coliforms are bacteria that are naturally present in the environment and are used as an indicator that other, potentially harmful, bacteria may be present. Total coliforms are common in the environment and are generally not harmful themselves. The presence of these bacteria can indicate that the water may have been contaminated with organisms that can cause disease. Some symptoms may include diarrhea, cramps, nausea, and possible jaundice, headaches and fatigue. When the monthly samples confirmed the presence of total coliform bacteria, we took steps to identify and correct the problem. Subsequent monthly sampling has confirmed the absence of total coliforms in the water system

The Drinking Water Source Protection Plan for Vineyard City Water System is available for your review. It contains information about source protection zones, potential contamination sources

and management strategies to protect our drinking water. Our sources have been determined to have a medium level of susceptibility from potential contamination sources such as septic tanks, roads, residential areas, industrial areas, etc. We have also developed management strategies to further protect our sources from contamination. Please contact us if you have questions or concerns about our source protection plan.

There are many connections to our water distribution system. When connections are properly installed and maintained, the concerns are very minimal. However, unapproved and improper piping changes or connections can adversely affect not only the availability, but also the quality of the water. A cross connection may let polluted water or even chemicals mingle into the water supply system when not properly protected. This not only compromises the water quality but can also affect your health. So, what can you do? Do not make or allow improper connections at your homes. Even that unprotected garden hose lying in the puddle next to the driveway is a cross connection. The unprotected lawn sprinkler system after you have fertilized or sprayed is also a cross connection. When the cross connection is allowed to exist at your home, it will affect you and your family first. If you'd like to learn more about helping to protect the quality of our water, call us for further information about ways you can help.

The Vineyard City Water System routinely monitors contaminants in our drinking water in accordance with the Federal and Utah State laws. The following table shows the results of our monitoring for 2024. It is important to remember that all water sources of drinking water contain some naturally occurring contaminants. At low levels, these substances are generally not harmful in our drinking water. Removing all contaminants would be extremely expensive, and in most cases, would not provide increased protection of public health.

In the following table you will find many terms and abbreviations you might not be familiar with. To help you better understand these terms we've provided the following definitions:

***Non-Detects (ND)*** - laboratory analysis indicates that the constituent is not present.

***ND/Low - High*** - For water systems that have multiple sources of water, the Utah Division of Drinking Water has given water systems the option of listing the test results of the constituents in one table, instead of multiple tables. To accomplish this, the lowest and highest values detected in the multiple sources are recorded in the same space in the report table.

***Parts per million (ppm) or Milligrams per liter (mg/l)*** - one part per million corresponds to one minute in two years or a single penny in $10,000.

***Parts per billion (ppb) or Micrograms per liter (ug/l)*** - one part per billion corresponds to one minute in 2,000 years, or a single penny in $10,000,000.

***Parts per trillion (ppt) or Nanograms per liter (nanograms/l)*** - one part per trillion corresponds to one minute in 2,000,000 years, or a single penny in $10,000,000,000.

***Parts per quadrillion (ppq) or Picograms per liter (picograms/l)*** - one part per quadrillion corresponds to one minute in 2,000,000,000 years or one penny in $10,000,000,000,000.

***Picocuries per liter (pCi/L)*** - picocuries per liter is a measure of the radioactivity in water.

***Millirems per year (mrem/yr)*** - measure of radiation absorbed by the body.

***Million Fibers per Liter (MFL)*** - million fibers per liter is a measure of the presence of asbestos fibers that are longer than 10 micrometers.

***Nephelometric Turbidity Unit (NTU)*** - nephelometric turbidity unit is a measure of the clarity of water. Turbidity in excess of 5 NTU is just noticeable to the average person.

***Action Level (AL)*** - the concentration of a contaminant which, if exceeded, triggers treatment or other requirements which a water system must follow.

***Treatment Technique (TT)*** - A treatment technique is a required process intended to reduce the level of a contaminant in drinking water.

***Maximum Contaminant Level (MCL)*** - The "Maximum Allowed" (MCL) is the highest level of a contaminant that is allowed in drinking water. MCLs are set as close to the MCLGs as feasible using the best available treatment technology.

***Maximum Contaminant Level Goal (MCLG)*** - The "Goal" (MCLG) is the level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety.

***Maximum Residual Disinfectant Level (MRDL)*** - The highest level of a disinfectant allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

***Maximum Residual Disinfectant Level Goal (MRDLG)*** - The level of a drinking water disinfectant below which there is no known or expected risk to health. MRDLGs do not reflect the benefits of the use of disinfectants to control microbial contaminants.

***Date***- Because of required sampling time frames i.e. yearly, 3 years, 4 years and 6 years, sampling dates may seem outdated.

***Waivers (W)***- Because some chemicals are not used or stored in areas around drinking water sources, some water systems have been given waivers that exempt them from having to take certain chemical samples, these waivers are also tied to Drinking Water Source Protection Plans.

| TEST RESULTS | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Contaminant** | **Violation Y/N** | **Level Detected ND/Low-High** | **Unit Measurement** | **MCLG** | **MCL** | **Date Sampled** | **Likely Source of Contamination** |
| **Microbiological Contaminants** | | | | | | | |
| Total Coliform Bacteria | N | 3 | N/A | 0 | Presence of coliform bacteria in 5% of monthly samples | 2024 | Naturally present in the environment |
| Fecal coliform and *E.coli* | N | ND | N/A | 0 | If a routine sample and repeat sample are total coliform positive, and one is also fecal coliform or *E. coli* positive | 2024 | Human and animal fecal waste |
| Turbidity for Surface Water | N | 0.51 | NTU | N/A | 0.5 in at least 95% of the samples and must never exceed 5.0 | 2024 | Soil Runoff<br><br>(highest single measurement & the lowest monthly percentage of samples meeting the turbidity limits) |
| **Inorganic Contaminants** | | | | | | | |
| Arsenic | N | 1-2 | ppb | 0 | 10 | 2024 | Erosion of natural deposits; runoff from orchards; runoff from glass and electronics production wastes |
| Barium | N | 87-96 | ppb | 2000 | 2000 | 2024 | Discharge of drilling wastes; discharge from metal refineries; erosion of natural deposits |
| Carbon, Total Organic (TOC) | N | 2-4 | ppm | NA | TT | 2024 | Naturally present in the environment |
| Copper<br>a. 90% results<br>b. # of sites that exceed the **AL** | N | a.0.556<br><br>b.0 | ppm | 1.3 | AL=1.3 | 2023 | Corrosion of household plumbing systems; erosion of natural deposits |
| Cyanide | N | ND-5 | ppb | 200 | 200 | 2024 | Discharge from steel/metal factories; discharge from plastic and fertilizer factories |
| Fluoride | N | 20-205 | ppb | 4000 | 4000 | 2024 | Erosion of natural deposits; water additive which promotes strong teeth; discharge from fertilizer and aluminum factories |
| Lead<br>a. 90% results<br>b. # of sites that exceed the **AL** | N | a. 5<br><br>b.0 | ppb | 0 | AL=15 | 2023 | Corrosion of household plumbing systems, erosion of natural deposits |

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|---|---|---|---|---|---|---|---|
| Selenium | N | 4 | ppb | 50 | 50 | 2024 | Discharge from petroleum and metal refineries; erosion of natural deposits; discharge from mines |
| Sodium | N | 7-35 | ppm | None set by EPA | None set by EPA | 2024 | Erosion of natural deposits; discharge from refineries and factories; runoff from landfills. |
| Sulfate | N | 9-64 | ppm | 1000 | 1000 | 2024 | Erosion of natural deposits; discharge from refineries and factories; runoff from landfills, runoff from cropland |
| TDS (Total Dissolved solids) | N | 112-444 | ppm | 2000 | 2000 | 2024 | Erosion of natural deposits |
| **Disinfection By-products** | | | | | | | |
| TTHM Total trihalomethanes | N | 14 | ppb | 0 | 80 | 2024 | By-product of drinking water disinfection |
| Haloacetic Acids | N | 6-7 | ppb | 0 | 60 | 2024 | By-product of drinking water disinfection |
| **Radioactive Contaminants** | | | | | | | |
| Alpha emitters | N | 4 | pCi/1 | 0 | 15 | 2024 | Erosion of natural deposits |
| Radium 228 | N | 0.62-1 | pCi/1 | 0 | 5 | 2024 | Erosion of natural deposits |

**WHY ARE THERE CONTAMINANTS IN MY DRINKING WATER?**

Drinking water, including bottled water, may reasonably be expected to contain at least a small amount of some contaminates. The presence of contaminants does not necessarily indicate that the water poses a health risk. More information about contaminates and the potential health effects can be obtained by calling the Environmental protection Agency's (EPA) Safe Drinking Water Hotline at (800-426-4791). The sources of drinking water (both tap and bottled water) include rivers, lakes, streams, reservoirs, springs and wells. As water travels over the surface of the land or through the ground, it dissolves naturally occurring minerals and, in some cases, radioactive material, and can pick up substances resulting from the presence of animals or from human activity: microbial contaminates, such as viruses and bacteria, that may come from sewage treatment plants, septic systems, agricultural livestock operations and wildlife; inorganic contaminants, such as salts and metals, which can be naturally occurring or result from urban stormwater runoff, industrial, or domestic wastewater discharges, oil and gas production, mining, or farming; pesticides and herbicides, which may come from a variety of sources such as agricultural, urban stormwater runoff, and residential uses; organic chemical contaminants, including synthetic and volatile organic chemicals, which are by-products of industrial processes and petroleum production, and can also come from gas stations, urban stormwater runoff, and septic systems; and radioactive contaminants, which can be naturally occurring or be the result of oil and gas production and mining activities. In order to ensure that tap water is safe to drink, EPA

prescribes regulations that limit the amount of certain contaminants in water provided by public water systems. Food and Drug Administration (FDA) regulations establish limits for contaminants in bottled water which must provide the same protection for public health.

If present, lead can cause serious health problems, especially for pregnant women and young children. The Vineyard City Water System has conducted thirty (30) lead samples during 2023. Sampling results can be obtained by calling Devan Peterson on 385-207-3893.

The Vineyard City Water System is responsible for providing high quality drinking water but cannot control the variety of materials used in plumbing components. Lead in drinking water is primarily from material and components associated with service lines and home plumbing. You share the responsibility for protecting yourself and your family from the lead in your home plumbing. You can take responsibility by identifying and removing lead materials within your home plumbing and taking steps to reduce your family's risk. If your water has been sitting for several hours, you can minimize the potential for lead exposure by flushing your tap for 30 seconds to 2 minutes before using water for drinking or cooking. You can also use a filter certified by an American National Standards Institute accredited certifier to reduce lead in drinking water. Lead in drinking water is rarely the sole cause of lead poisoning, but it can add to a person's total lead exposure. All potential sources of lead in the household should be identified and removed, replaced or reduced. If you are concerned about lead in your water, you may wish to have your water tested. Please contact Vineyard City Water System at Devan Peterson on 385-207-3893. Information on lead in drinking water, testing methods, and steps you can take to minimize exposure is available at *http://www.epa.gov/safewater/lead*.

Some people may be more vulnerable to contaminants in drinking water than the general population. Immunocompromised people such as people with cancer undergoing chemotherapy, people who have undergone organ transplants, people with HIV/AIDS or other immune system disorders, some elderly, and infants can be particularly at risk from infections. These people should seek advice from their health care providers about drinking water. EPA/CDC guidelines on appropriate means to lessen the risk of infection by cryptosporidium and other microbiological contaminants are available from the Safe Drinking Water Hotline (800-426-4791).

We at The Vineyard City Water System work around the clock to provide top quality water to every tap. We ask that all our customers help us protect our water sources, which are the heart of our community, our way of life and our children’s future.